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Scandinavian Journal of Medicine & Science in Sports

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ORIGINAL ARTICLE OPEN ACCESS

Blood, Sweat, and Tears: Implications for Hydration Testing in Combat Sports—Investigating Body Mass Loss and Biomarker Changes

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Received: 16 March 2024 | **Revised:** 13 June 2024 | **Accepted:** 10 July 2024

Funding: This work was supported by a research grant from the Western Australia Combat Sports Commission, which supports CSD's PhD scholarship. This research was also supported by an Australian Government Research Training Program Scholarship.

Keywords: blood | dehydration | passive heating | physiology | reproducibility | urine

ABSTRACT

Combat sports athletes often undergo rapid body mass loss (BML), which presents health risks. Hydration testing has been proposed as a possible solution to reduce or eliminate rapid BML. However, combat sports athletes may exhibit distinct physiological characteristics due to repeated exposure to BML. Thus, traditional and emerging hydration biomarkers should be investigated to determine their potential suitability for field use in this cohort. This study examined whether BML can explain changes in serum and urine osmolality (SosmΔ, UosmΔ), tear osmolarity (TosmΔ), hematocrit (HctΔ), and urine-specific gravity (USGΔ) after mild–moderate passive dehydration. Biomarker reliability was also assessed across two trials. Fifteen male and female combat sports athletes (age: 26.3 ± 5.3 years, body mass: 67.7 ± 9.9 kg) underwent a sauna protocol twice (5–28 days apart) aiming for 4% BML. The average BML in Trials 1 and 2 was 3.0 ± 0.7%. Regression analysis revealed that BML explained HctΔ ($R^2 = 0.22$, $p = 0.009$) but not SosmΔ ($R^2 = 0.11$, $p = 0.079$) or other biomarkers. Intraclass correlation coefficients (ICCs) were significant for all biomarkers except TosmΔ (ICC = 0.06, $p = 0.37$) and post-Tosm (ICC = 0.04, $p = 0.42$); post-Hct performed best (ICC = 0.82, $p < 0.001$). Contingency tables with post-Sosm (295 mOsm/kg) and post-USG (1.020) cutoffs revealed an 80% true negative rate (TNR) and a 62% true positive rate (TPR). Increasing the Sosm cutoff to 301 mOsm/kg decreased the TNR to 52% but increased the TPR to 83%. Although blood parameters were most sensitive to BML, they could only explain 11%–22% of biomarker variation. The typical USG cutoff misclassified 42% of athletes postdehydration, and reliability was generally poor-moderate. Alternative strategies should be pursued to manage rapid BML in combat sports.

1 | Introduction

In combat sports, athletes compete by weight class, where body mass (BM) is measured at an official weigh-in on the day of or before a competition. Hydration assessment is of interest because athletes often rapidly reduce their total body water (TBW) before the weigh-in [1]. Previous research reported that within 24 h of weigh-ins, Brazilian jiu-jitsu (BJJ) and mixed martial arts (MMA) athletes reduce BM by, on average, ~1.4 and ~3.4 kg, respectively [1]. Dehydration via passive heat exposure is a common rapid body mass loss (BML) method during this period, especially in MMA athletes, where ~76% of individuals report sauna use [1]. However, prolonged heat exposure and extreme dehydration can pose serious health risks. Previous studies have reported acute kidney damage [2], suppressed immune function [3], and in rare cases, death [3]. In efforts to mitigate extreme BML practices, certain sports like wrestling introduced hydration testing in 1998 [4]. This testing has been an integral

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component of their minimum weight certification program ever since. One aspect of the program involves evaluating athletes' hydration status in the preseason using urine-specific gravity (USG). If an athlete's USG exceeds 1.020, they are unable to proceed with the assessment at that time, as it could potentially affect body composition results [4]. However, in recent years, the use of urine tests to assess hydration status in other combat sports organizations has raised concerns [5–9]. In particular, whether rapid BML can accurately explain the changes in hydration biomarkers, and whether these results are reliable [10–15]. Furthermore, there is a misclassification cost [7], as it may result in unjust disqualification or inclusion of competitors, potentially negatively affecting their health, careers, and finances. To date the literature presents conflicting findings on the accuracy of hydration testing, highlighting the need for further investigation [10–15].

Previous research has explored various biomarkers of hydration status in exercise-induced dehydration contexts [11, 16–20]. However, exercise itself may influence biomarker testing. Factors such as metabolic water release from muscle glycogen, increased respiration, reduced blood flow to the kidneys, and increased metabolism could impact biomarker changes [21]. In contrast, metabolic water production might counterbalance respiratory water loss, and the water associated with muscle glycogen might not significantly impact fluid balance or dehydration [22]. However, despite these contrasting viewpoints, previous research has shown differing biomarker responses between passive and active dehydration [20]. Moreover, some authors have speculated that combat sports athletes may have adaptive mechanisms that protect them from the adverse effects of rapid BML [23], and thus may display distinct physiological changes that could influence biomarker testing. Therefore, considering the widespread use of saunas [1], and the potentially unique physiological response to dehydration, it is important to investigate whether BML can accurately explain changes in hydration biomarkers following sauna-induced dehydration in this cohort.

Despite limitations [24], serum osmolality (Sosm) is widely regarded as the primary marker for acute dehydration assessment under controlled laboratory conditions, particularly when the BM remains stable [25, 26]. The accuracy of both urine osmolality (Uosm) and USG as hydration indicators has been supported [17, 27] and questioned [8, 15, 28]. Tear osmolarity (Tosm), although showing a robust correlation with Sosm, exhibits notable individual variation and remains a relatively unexplored biomarker [29]. Similarly, hematocrit (Hct) varies by 5%–6% between individuals and 2%–3% within individuals [30], yet research on Hct change scores, which are potentially more suitable than spot measures, is limited [5]. Notably, no previous studies have investigated the suitability of multiple hydration biomarkers following sauna-induced dehydration among male and female combat sports athletes. This study aimed to fill this research gap by examining the accuracy of BML in explaining changes in hydration biomarkers (Sosm, Tosm, Uosm, Hct, and USG) following mild–moderate acute dehydration. Additionally, we assessed the reliability of these biomarkers across two repeat sessions with the same participants. Contingency tables are used to determine if postdehydration USG scores accurately classify athletes as hydrated or dehydrated based on postdehydration Sosm values. We hypothesize that BML will explain a moderate proportion of biomarker changes, with Sosm showing the strongest association. Furthermore, we anticipate large variations in the reliability of all biomarkers across trials, with Sosm demonstrating the most consistent results. We also expect postdehydration USG cutoffs to misclassify a large proportion of athletes. Our findings will provide insight into whether any biomarker is promising for hydration testing in the field and if they are reliable. Since certain organizations use USG to assess athletes' hydration status, the findings from this analysis will have practical implications for ongoing hydration testing in the field. Overall, our research will provide valuable information for organizations that are already conducting hydration testing at weigh-ins, as well as for those contemplating its implementation to discourage rapid dehydration.

2 | Methods

2.1 | Ethics Approval

This study received approval from the Edith Cowan University Human Research Ethics Committee (2022-03382-DOHERTY). All individuals were fully informed about the experimental procedures and risks before participating, and informed consent was provided.

2.2 | Participants

Fifteen male and fifteen female combat sports athletes (age 26.3 ± 5.3 years, body mass 67.7 ± 9.9 kg) were recruited for this study through social media platforms, local gyms, and sporting events across Perth from October 22, 2022, to April 23, 2023. The athletes represented different combat sports, including Muay Thai ($n = 14$), MMA ($n = 6$), BJJ ($n = 6$), and boxing ($n = 4$). The sample contained experienced national champions and athletes with limited competitive experience. We intentionally included a range of experience levels to best represent the combat sports population. Before participating in the study, all participants completed a comprehensive medical history questionnaire to ensure that they had no contraindications for testing. Participants were excluded if they wore contact lenses that could affect their Tosm measurements [31], or if they were taking medication known to interfere with fluid balance, except for contraceptives. In practice, hydration testing is not based on the phase of the menstrual cycle, and research suggests that the menstrual stage does not significantly influence volume regulatory responses [32]. Therefore, females were included and studied regardless of their current cycle phase.

2.3 | Study Protocol

The participants underwent a pre- and post-sauna hydration assessment, aiming to achieve a 4% reduction in BM (Figure 1). This 4% reduction was selected with ethics considerations in mind, as well as in alignment with reported BML (1.5%–4%) in combat sports athletes 24 h before weigh-ins [1]. The protocol was repeated twice by all 30 participants within 5–28 days (average 10.9 ± 5.6) of Trial 1 (T1). Before the trial, participants were instructed to drink 45 mL of fluid per kilogram of body mass

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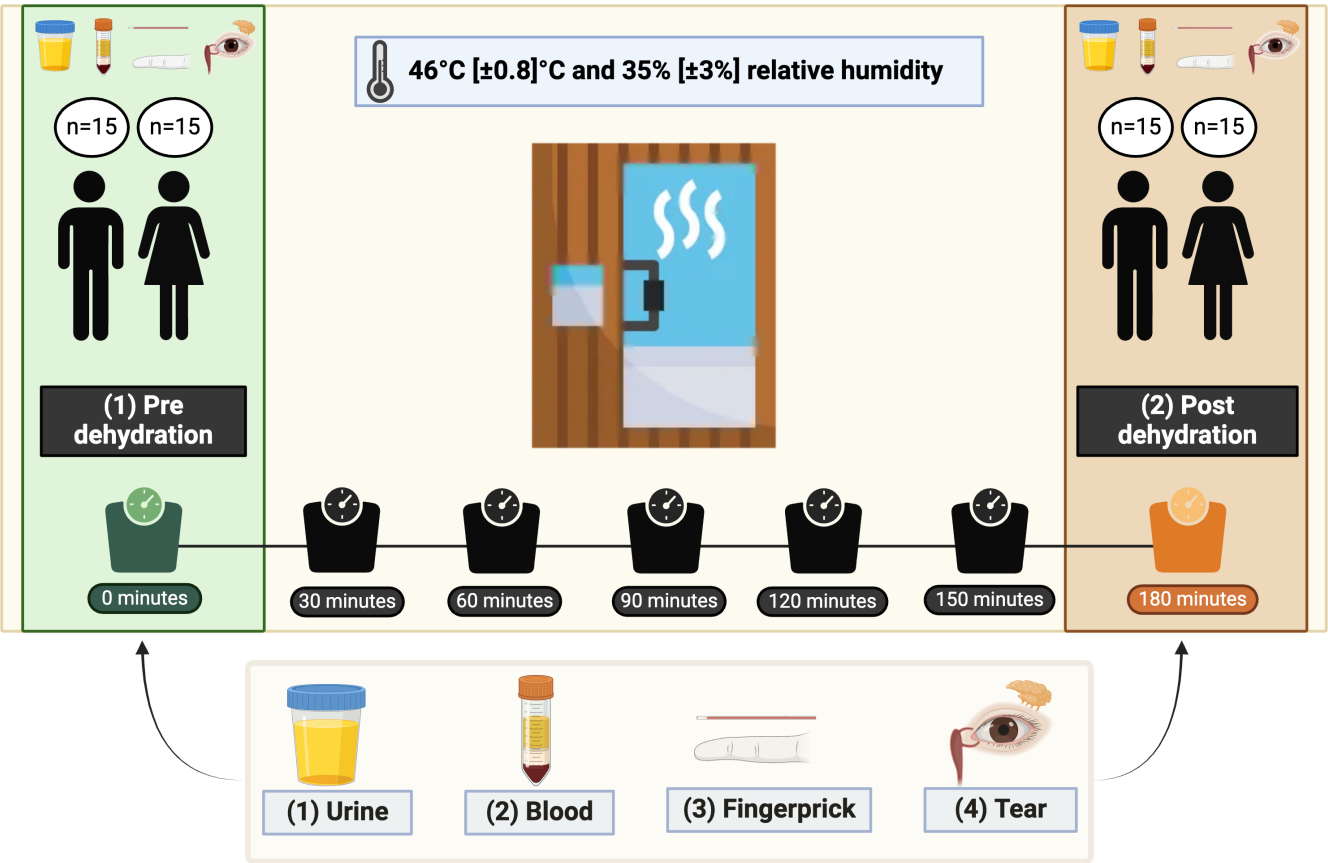

FIGURE 1 | Overview of the sampling and dehydration protocol. Hydration biomarkers were assessed pre- and post-dehydration. Body mass was collected at all time points from 0 to 180 min to determine body mass loss. Biomarkers were collected in the order they appear in the figure.

for 2 days and maintain their regular diet. They were advised to avoid intense and heavy contact exercise, and alcohol for 24 h before testing. On the testing day, participants arrived after fasting for ~12 h. Food and fluid intake were recorded the day before T1 for replication before Trial 2 (T2) but were not reported to the research team. Participants were given a urine color chart to monitor their fluid intake and were asked to consume 500 mL of water 1 h before testing on both days (Supplementary File 1). Upon arrival at the laboratory between 7:00 and 9:30 a.m., participants provided a midflow urine sample. Hydration status was assessed by USG, with a value below 1.020 indicating euhydration, and a value equal to or above 1.020 indicating hypohydration. Only one out of 60 visits showed a USG above 1.020, and the participant was given 500 mL of water in 15-min intervals until an acceptable measure was obtained (total = 1500 mL). Hydration markers were collected and analyzed in the order of urine, blood, and tear.

2.4 | Dehydration Protocol

Both T1 and T2 were performed at a similar time of day (0.1 ± 0.2 h). Passive heat exposure (46°C [±0.8]°C and 35% [±3%] relative humidity) involved a maximum of six 30-min sessions using an infrared sauna (3 Person Infra-Red Hemlock Sauna; West Coast Spas, Joondalup, Australia). Every 30 min, participants had a 5–10-min rest period, during which time the nude BM was measured (Figure 1). If a 4% BM reduction was achieved before the six 30-min sessions were completed (13/60), heat exposure was stopped, and the postsauna hydration assessment started.

2.5 | Hydration Measures

Midflow urine samples were collected upon arrival, USG (Atago hand refractometer, model UNC-NE; Atago, Minato, Japan) was assessed by refractometry, and Uosm was measured by freezing point depression osmometry (Advanced 3250 single-sample osmometer; Advanced Instruments, Norwood, Canada). Venous blood samples were collected by using serum-separating tubes after participants had been seated for 3–5 min. The samples were then centrifuged at 12 000 rpm for 10 min at 4°C (Multifuge 3 S-R; Kendro, Asheville, NC, USA). The serum was separated, and osmolality (Advanced 3250 single-sample osmometer; Advanced Instruments) was assessed. Next, fingerprick blood samples were collected using capillary tubes (Capilette MPW212; MPW Med. Instruments, Warsaw, Poland) and centrifuged at 3500 rpm for 5 min to determine the hematocrit percentage (Centrifuge MPW212; MPWMed. Instruments).

Tear samples were collected and analyzed using a tear osmometer (TearLab Osmolarity System; TearLab, San Diego, CA, USA). Participants were asked to blink three times and squeeze their

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eyes shut, producing fresh tear fluid from the lacrimal gland [31]. Tear fluid was then collected from the left eye. The pen audibly beeped once a sufficient volume (~50 nL) was obtained. It was immediately docked on the osmolarity system reader, and within 10 s, the tear osmolarity value was presented.

2.6 | Statistical Approach and Analysis

The statistical approach used in this study, which involves associations, regression coefficients, and Bland–Altman plots, serves unique and complementary purposes. Associations are valuable for identifying relationships between BML and hydration biomarkers, showing how these variables change in relation to each other. Regression coefficients further enhance our understanding by quantifying the strength and direction of these relationships, helping to determine whether BML can explain a significant portion of the variation in hydration biomarkers. Finally, Bland–Altman plots were used to assess biomarker agreement between T1 and T2, detecting any systematic biases that may exist. Although these methods may overlap to some extent, they collectively provide comprehensive insight into the performance of hydration tests, improving the robustness and reliability of our findings. This multifaceted approach is standard in scientific research, with each method offering a distinct perspective.

To assess data normality and homogeneity, Shapiro–Wilk's and Levene's tests were used. All data are reported as the mean ± standard deviation (SD) unless otherwise specified. Given the non-normal distribution of the BML variable, Spearman's rank correlation coefficient (R_S) was used to examine the relationships. Because of heteroscedasticity and non-normal residuals observed in mixed models, a simple linear regression was conducted using T1 data to investigate the relationships between BML and biomarkers (HctΔ, SosmΔ, TosmΔ, USGΔ, and UosmΔ), which did not violate the model assumptions. Attempts were made to address heteroscedasticity using robust and generalized linear mixed model regression, but model violations persisted. We chose to predict biomarker changes using BML rather than the reverse configuration to ensure a consistent scale for interpretation. Utilizing blood, urine, and tear biomarkers to predict BML presents challenges because a one-unit increase in each biomarker does not carry the same meaning, due to variations in their scales and physiological interpretations. Additionally, the regression output for BML predicting biomarker changes provides specific changes in each biomarker, regardless of its scale. Notably, irrespective of predictor and outcome configurations, the models consistently yielded identical p values and coefficient of determination (R^2) values. Fixed effects such as sex, sauna time, and sweat rate were intentionally excluded from the analysis, as they are not adjusted or accounted for in practice. Model accuracy was evaluated using the standard error of the estimate (SEE) and residual mean square error (RMSE). Additionally, the R^2 was used to assess model fit and determine the proportion of variance in each biomarker explained by BML.

To assess test–retest reliability, we compared pre, post, and biomarker change scores between T1 and T2. Our reliability analysis included the intraclass correlation coefficient (ICC), coefficient of variation (CV), and SEE. For the ICC, we used the ICC 2,1 model, treating the rater as fixed and measurements (T1, T2) as random, following the approach of Koo and Li (2016) for categorization: poor (<0.50), moderate (0.50–0.75), good (0.75–0.90), or excellent (>0.90) [33]. The CV, quantifying relative variability among participants, was calculated individually for T1 and T2, with negative-valued variables excluded. Though not a conventional method for employing this statistic, the aim was to discern whether the variability among participants at T1 and T2 was comparable. The formula used for each trial and variable was as follows:

$$\mathrm{CV} = \left(\frac{\mathrm{SD}}{\mathrm{Mean}}\right) \times 100$$

Notably, the CV can vary substantially based on the unit of measurement. For instance, consider the predehydration Uosm mean ± SD (344 ± 202) with that of Sosm (287 ± 4). In the case of Uosm, a larger SD indicates a wider range of values (min–max), resulting in a larger CV. Thus, it is more informative to compare, for example, Uosm with Uosm across T1 and T2 rather than between Uosm and Sosm within trials. This approach offers insights into trial, and biomarker-specific variability, enabling us to assess whether relative reliability remained consistent across trials, even when absolute variability was high. For each trial and variable, the SEE was computed using the following formula:

$$\mathrm{SEE} = \mathrm{SD} * \sqrt{1 - \mathrm{ICC}}$$

Bland–Altman plots were used to assess trial agreement, and linear regression was used to analyze the proportional bias between T1 and T2. The limits of agreement (LOA) were determined by:

$$\mathrm{LOA} = \mathrm{mean(differences)} \pm 1.96 * \mathrm{SD(differences)}$$

Hydration status classification was evaluated using contingency tables, employing commonly used cutoffs from prior hydration research [5, 25]. Sensitivity and specificity were assessed, with false positives and negatives determined based on post-Sosm and post-USG values from both trials ($n = 60$). These markers were selected because Sosm is considered the most reliable biomarker of hydration status, while USG is commonly employed in field and regulatory settings [7, 25]. A false negative was recorded when a participant had a Sosm ≥295 mOsm/kg and a USG <1.020. Conversely, a false positive was indicated when Sosm was <295 mOsm/kg and USG was ≥1.020. A separate matrix was created to assess a Sosm cutoff of 301 mOsm/kg [26].

Previous regression analyses reported correlation coefficients (R) rather than R^2 and were insufficient to determine the effect size for power calculations [10, 34]. To address this, we used a conservative effect size of 0.3 for all biomarkers using linear regression with one predictor (BML), a significance level of 0.05, and a minimum desired power of 0.8. Using G*Power version 3.1.9.6, we estimated the sample size for a two-tailed

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test to be 28 participants. For the ICC analysis, sample size calculation was conducted using the R package "pwr," which indicated a need for 29 participants to achieve a power of 0.8, an alpha of 0.05, and a target ICC value of 0.5. Statistical analyses were carried out using R Studio (version 4.2.1, R Foundation for Statistical Computing, Vienna, Austria), with statistical significance set at $p < 0.05$. To prepare, visualize, and analyze the data, we used several packages, including *lme4, car, performance, dlookr, stats, gtsummary, ggplot, GGally, tidyverse, psych, BlandAltmanLeh, and irr.* Biorender.com was also used to create Figure 1.

3 | Results

All 30 participants completed both sessions. The participants had similar BMLs at T1 and T2 (Table 1). There was no significant difference in any of the biomarkers at predehydration or postdehydration between T1 and T2 ($p < 0.05$, data not shown).

3.1 | Regression Analyses

The full regression analyses are presented in Table 2. In Model 1 (formula: HctΔ ~ BML), the effect of BML was found to be statistically significant and positive ($t(28/30) = 2.81$, $p = 0.009$; Std. beta = 0.47, 95% CI [0.13, 0.81]) with an RMSE of 1.2%. However, in Model 2 (formula: UosmΔ ~ BML), the effect of BML was statistically nonsignificant and positive ($t(28/30) = 0.50$, $p = 0.622$; Std. beta = 0.09, 95% CI [−0.29, 0.48]), with an RMSE of 301 mOsm/kg. Similarly, in Model 3 (formula: USGΔ ~ BML), the effect of BML was statistically nonsignificant and positive ($t(28/30) = 0.77$, $p = 0.450$; Std. beta = 0.14, 95% CI [−0.24, 0.53]), with an RMSE of 0.008 units. Model 4 (formula: TosmΔ ~ BML) also yielded a statistically nonsignificant and positive effect of BML ($t(28/30) = 1.51$, $p = 0.142$; Std. beta = 0.27, 95% CI [−0.10, 0.65]), with an RMSE of 14 mOsm/L. In Model 5 (formula: SosmΔ ~ BML), the effect of BML was found to be statistically nonsignificant and positive ($t(28/30) = 1.8$, $p = 0.079$; Std. beta = 0.33, 95% CI [−0.04, 0.69]), with an RMSE of 4 mOsm/kg. Figure 2 visually represents the T1 data and the predehydration biomarker values at 0% BML. Additionally, the correlation data and regression line are presented. Only post-Tosm and post-Hct were significantly correlated with BML ($p < 0.05$).

3.2 | Test–Retest Reliability

Table 3 displays the test–retest relative reliability analysis for pre- and post-dehydration and biomarker changes. All variables showed a significant ICC ($p < 0.05$) between T1 and T2, except for TosmΔ and post-Tosm. Figure 3 displays the Bland–Altman plots for BML and each biomarker. The LOA and SD were generally large, with relatively low mean differences. The BML, USGΔ, HctΔ, SosmΔ, and UosmΔ regression lines showed no systematic bias between T1 and T2 ($p > 0.05$), but the TosmΔ plot (Figure 3D) indicated a positive proportional bias ($p = 0.048$). The CVs within variables across trials were consistent, although the absolute values for each trial were greatest for change scores and lower for pre- and post-measurements.

TABLE 1 | Summary of the pre- and post-dehydration hydration indices for Trials 1 and 2.

Characteristic	Trial 1 ($n = 30$)			Trial 2 ($n = 30$)		
	PRE-DHY	POST-DHY	Difference [95% CI]	PRE-DHY	POST-DHY	Difference [95% CI]
Body mass, kg	69.6 ± 10.0 (54.6–88.4)	67.5 ± 9.5 (53.4–85.3)	2.13 [1.88, 2.39]	69.9 ± 10.0 (55.2–87.3)	67.8 ± 9.6 (53.7–84.9)	2.08 [1.85, 2.32]
BML, %	—	3.0 ± 0.7 (2.1–4.3)	—	—	3.0 ± 0.7 (2.0–4.1)	—
Serum osmolality, mOsm/kg	287 ± 4 (279–298)	299 ± 4 (292–310)	−12.5 [−14.0, −11.0]	288 ± 4 (280–299)	299 ± 4 (293–310)	−10.5 [−11.9, −9.21]
Urine osmolality, mOsm/kg	344 ± 202 (64–784)	712 ± 238 (276–1093)	−368 [−479, −257]	421 ± 237 (107–891)	748 ± 235 (371–1196)	−327 [−443, −211]
Tear osmolarity, mOsm/L	294 ± 7 (277–306)	306 ± 11 (291–334)	−11.5 [−16.7, −6.32]	293 ± 7 (282–309)	304 ± 10 (283–322)	−11.7 [−15.2, −8.11]
Hematocrit, %	41 ± 2 (37–45)	44 ± 3 (38–48)	−2.30 [−2.83, −1.77]	41 ± 2 (37–45)	43 ± 3 (37–48)	−2.10 [−2.57, −1.63]
Urine-specific gravity	1.009 ± 0.006 (1.001–1.019)	1.021 ± 0.007 (1.008–1.033)	−0.012 [−0.015, −0.009]	1.011 ± 0.006 (1.002–1.019)	1.021 ± 0.007 (1.011–1.033)	−0.010 [−0.013, −0.007]

Note: All values are the mean ± SD (min–max).
Abbreviations: BML, body mass loss; POST-DHY, postdehydration; PRE-DHY, predehydration.

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TABLE 2 | Linear regression and correlation analyses for Trial 1 data ($n = 30$) where body mass loss predicts hydration biomarker change.

Variable	Regression equation	R_S	R^2	Intercept (reference)	Slope (change)	Confidence interval	SEE (precision)	Model p values
Model 1 ($n = 30$)	y (HctΔ) = −0.6 + 1 * BML	0.49*	0.22	−0.6	—	−2.7 to 1.6	1.0	0.592
BML (%)	—	—	—	—	1	0.3 to 1.6	0.3	0.009**
Model 2 ($n = 30$)	y (UosmΔ) = 248 + 40 * BML	0.14	0.01	248	—	−258 to 753	247	0.324
BML (%)	—	—	—	—	40	−123 to 202	79	0.622
Model 3 ($n = 30$)	y (USGΔ) = 0.007–0.002 * BML	0.12	0.02	0.007	—	−0.007 to 0.020	0.007	0.305
BML (%)	—	—	—	—	−0.002	−0.003 to 0.006	0.002	0.450
Model 4 ($n = 30$)	y (TosmΔ) = −5 + 5 * BML	0.34	0.08	−5	—	−28 to 18	11	0.661
BML (%)	—	—	—	—	5	2 to 13	4	0.142
Model 5 ($n = 30$)	y (SosmΔ) = 6.8 + 2 * BML	0.29	0.11	6.8	—	0.1 to 13.4	3.2	0.043#
BML (%)	—	—	—	—	2.0	−0.2 to 4.0	1.0	0.079

Abbreviations: BML, body mass loss; Hct, hematocrit, R^2, coefficient of determination; R_S, spearman rank correlation coefficient; SEE, standard error of the estimate; Sosm, serum osmolality; Tosm, tear osmolality; Uosm, urine osmolality, USG, urine-specific gravity; Δ, change.
*Indicates significant Spearman correlation between BML and the biomarker ($p < 0.05$).
#Indicates the intercept was significantly different from 0 ($p < 0.05$).
**Indicates a significant relationship between dependent (biomarker) and independent variable (BML) ($p < 0.05$).

3.3 | Contingency Tables

In Table 4, the lower Sosm cutoff (295 mOsm/kg) demonstrated a greater ability to detect values below the specified thresholds (4/5, specificity), while the higher Sosm performed better in identifying true dehydration (15/18, sensitivity). Among the post-USG values, 25/60 samples had values <1.020. In comparison, 5/60 post-Sosm measurements were <295 mOsm/kg, and 42/60 post-Sosm measurements were <301 mOsm/kg.

4 | Discussion

The aim of this investigation was to explore whether BML could explain changes in hydration biomarkers and assess their test–retest reliability across two trials. This study encompassed several distinctive aspects: an examination of multiple biomarkers, including the understudied Tosm; the use of a dehydration method commonly employed by combat sports athletes; and an evaluation of both accuracy and reliability. Notably, we also included female participants, a group often underrepresented in combat sports and sports science research. For a hydration test to be considered for regulatory use, BML should be able to explain most of the variation in biomarker changes, irrespective of individual training regimens and dietary practices [7]. Although we observed notable differences in biomarkers, BML alone could not precisely account for biomarker changes. Moreover, most of our reliability results fell within the poor to moderate range. Consequently, under the conditions studied, we cannot recommend hydration testing as a viable solution for managing rapid BML in combat sports. This conclusion is consistent with prior research highlighting the limitations of other hydration biomarkers [6, 7, 9].

4.1 | Blood Measures

Importantly, while our approach aimed to address a scientific question and identify whether any biomarker may be suitable for future research and possible regulatory use, it is unrealistic to obtain blood values in the field. Therefore, the Sosm and Hct are intended to serve as a reference for the more practical urinary and tear measurements in this study. Nevertheless, a large portion of the variation in HctΔ and SosmΔ remained unexplained by BML. The RMSEs of 4 mOsm/kg and 1.2% indicate that the models had relatively high overall errors compared with the estimates (Table 2). Although the BML showed statistical significance in predicting HctΔ, it explained only 22% of the variation, indicating that factors outside BML contribute to changes in Hct. A higher Sosm dehydration cutoff of 301 mOsm/kg showed adequate sensitivity (>80%) for detecting dehydration in participants [26]. However, this, the LOAs for SosmΔ (−6.4 to 10.3 mOsm/kg) and HctΔ (−2.1 to 2.5%) were wide, as depicted in Figure 3. For new participants with similar attributes to those in the study, their test–retest differences should fall within these ranges. However, these LOAs suggest large variations in the differences between T1 and T2. Furthermore, although not feasible, if these biomarkers were studied under field conditions, an even larger variation is expected, where athletes dehydrate using various methods and to different magnitudes [1]. The test–retest reliability of the SosmΔ was poor (ICC < 0.5), while that of the HctΔ was moderate

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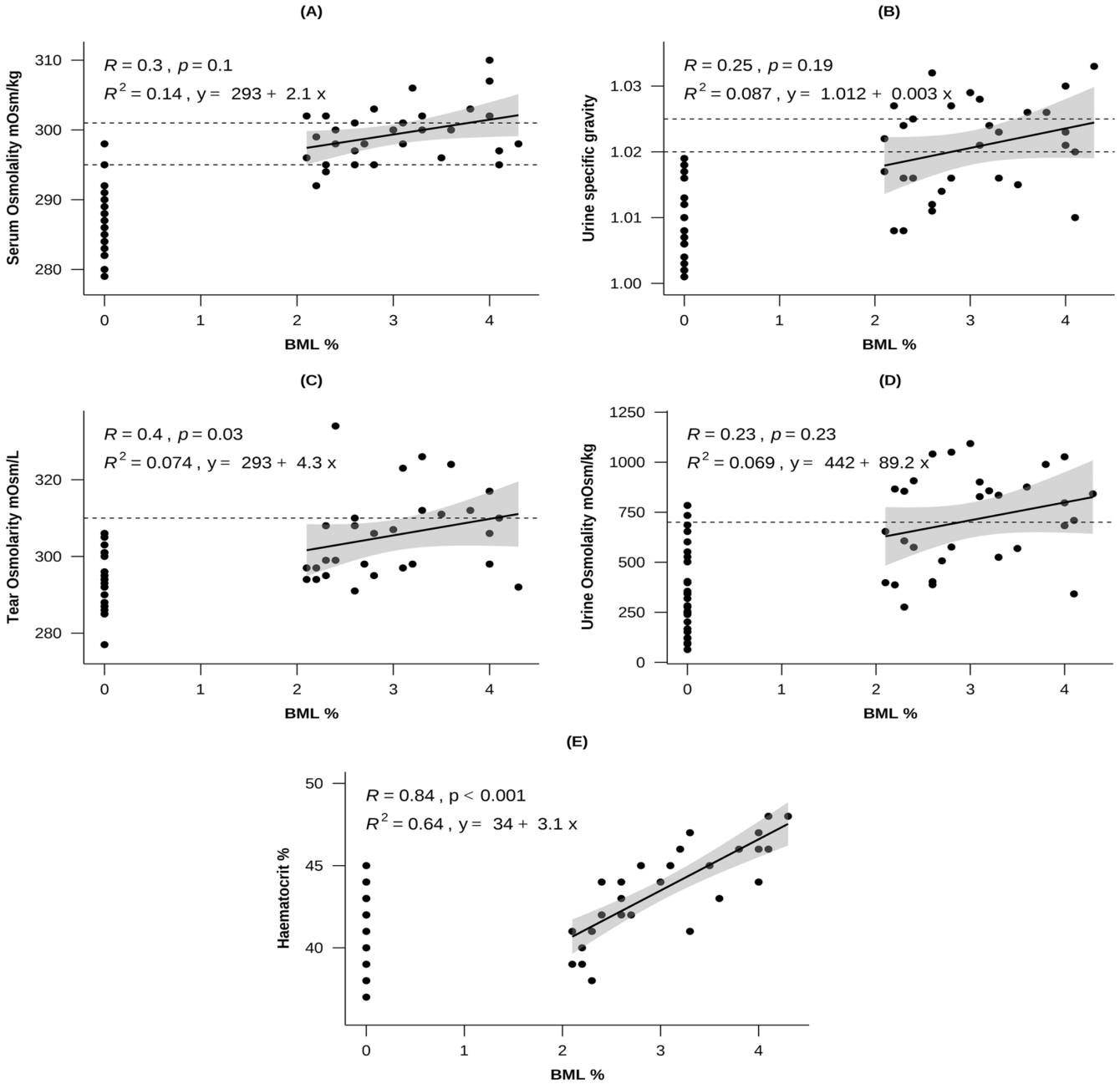

FIGURE 2 | Plots A–E depict regression results for the BML predicting postdehydration (*x*-axis, starting at 2.1%) biomarker scores (*y*-axis) from Trial 1. Data points at 0% BML represent predehydration values for all biomarkers but were not included in the regression analyses. The gray shaded area represents the standard error for regression. Spearman's correlation coefficient (R_S), *p* value, coefficient of determination (R^2), and regression equation are displayed for each plot. The dashed lines on the graphs represent the proposed dehydration cutoffs; for urine-specific gravity (1.020 and 1.025) and serum osmolality (295 and 301 mOsm/kg), two thresholds have been proposed [10, 25, 26], the urine osmolality threshold is 700 mOsm/ kg [11, 31], and tear osmolarity is 310 mOsm/L [31].

(ICC = 0.61). The pre- and post-measures of both Sosm and Hct exhibited relatively low and consistent SEE and CV across trials compared with the change scores. This suggests that the variation in HctΔ and SosmΔ between participants is greater than the variation in their pre- or post-values when the BML is between 2% and 4.3%. This discrepancy is likely due to different physiological responses to dehydration, and the change scores were lower absolute numbers compared with the pre- and post-dehydration values, resulting in more similar SDs and means. Compared with our findings, Ring et al. reported a greater ICC for Hct at rest [35]. However, information on the ICC model, type, and definition was not reported, making meaningful comparisons difficult. We found a significant correlation between BML and HctΔ, and between BML and post-Hct. This was not observed for the BML and Sosm measures, possibly due to the wider range of Sosm responses between individuals. For instance, 13/60 sessions achieved a BML of 4%–4.3%, but the SosmΔ ranged from 8 to 20 mOsm/kg, whereas for HctΔ, it was 2%–5%. It is possible that factors such as sweat sodium and heat acclimation status may have contributed to more variation in SosmΔ than in HctΔ [36]. Although Hct is susceptible

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TABLE 3 | Test–retest relative reliability analyses of hydration biomarkers between Trials 1 ($n = 30$) and 2 ($n = 30$).

Variable	ICC 2,1 (95% CI)	Test–Retest Reliability	*p* value	SEE (T1)	SEE (T2)	CV % (T1)	CV% (T2)
BML, %							
Change	0.80 (0.63 to 0.90)	Moderate-Good	<0.001*	0.3	0.3	23.2	23.5
Body mass, kg							
Change	0.90 (0.80 to 0.95)	Good-Excellent	<0.001*	0.2	0.2	31.8	30.2
Pre	0.99 (0.992 to 0.998)	Excellent	<0.001*	0.6	0.6	14.4	14.3
Post	0.99 (0.992 to 0.998)	Excellent	<0.001*	0.6	0.6	14.1	14.2
Serum osmolality, mOsm/kg							
Change	0.35 (0.02 to 0.62)	Poor-Moderate	0.019*	3.3	2.9	32.8	33.7
Pre	0.38 (0.04 to 0.64)	Poor-Moderate	0.014*	3.3	3.5	1.4	1.6
Post	0.49 (0.16 to 0.72)	Poor-Moderate	0.003*	2.9	2.8	1.4	1.3
Urine osmolality, mOsm/kg							
Change	0.60 (0.31 to 0.79)	Poor-Good	<0.001*	188	196	—	—
Pre	0.50 (0.18 to 0.72)	Poor-Moderate	0.002*	143	168	58.6	56.2
Post	0.47 (0.14 to 0.71)	Poor-Moderate	0.004*	174	171	33.4	31.4
Tear osmolarity, mOsm/L							
Change	0.06 (−0.32 to 0.41)	Poor	0.376	14	9	—	—
Pre	0.46 (0.11 to 0.69)	Poor-Moderate	0.006*	5	5	2.4	2.5
Post	0.04 (−0.3 to 0.40)	Poor	0.417	11	10	3.6	3.2
Hematocrit, %							
Change	0.61 (0.33 to 0.79)	Poor-Good	<0.001*	0.9	0.8	61.6	60.4
Pre	0.64 (0.36 to 0.81)	Poor-Good	<0.001*	1.3	1.5	5.0	6.1
Post	0.82 (0.66 to 0.91)	Moderate-Excellent	<0.001*	1.2	1.4	6.3	7.5
Urine-specific gravity							
Change	0.46 (0.14 to 0.70)	Poor-Moderate	0.003*	0.006	0.006	—	—
Pre	0.49 (0.18 to 0.72)	Poor-Moderate	0.002*	0.004	0.004	0.55	0.58
Post	0.42 (0.07 to 0.67)	Poor-Moderate	0.010*	0.005	0.005	0.70	0.65

Note: ICC 2,1 Model = "two-way", type = "agreement", unit = "single". An ICC less than 0.5 indicates *poor reliability*, values between 0.5 and 0.75 indicate *moderate reliability*, values between 0.75 and 0.9 indicate *good reliability*, and values greater than 0.90 indicate *excellent reliability*.
Abbreviations: BML, body mass loss; CV, coefficient of variation; ICC, intraclass correlation coefficient; SEE, standard error of the estimate; T1, Trial 1; T2, Trial 2.
*Indicates significant agreement (correlation) among the observed measures at T1 and T2 ($p < 0.05$).

to fluctuations due to sex differences [19], postural changes, body movement, and skin temperature [30], it performed better than Sosm. Previous research has also shown that a 500 mL water bolus administered pretrial can dilute Sosm by more than −3 mOsm/kg [37], and biological variation analysis indicates an intraindividual difference of 2–4 mOsm/kg [30]. These factors combined likely explain most of the variation and poor explanatory power of BML for the Sosm and Hct measures.

Several investigations have shown that Sosm is sensitive to changes in hydration status based on mean comparisons [15, 17, 38]. However, mean changes only explain the differences in two variables at the group level. Figure 2 displays the R_S, *p* value, R^2, and regression equation for BML predicting the postdehydration biomarker scores in T1. The correlation between BML and post-Sosm was found to be $R_S = 0.3$, but the proportion of variance explained by BML was less than half of the correlation ($R^2 = 0.14$). Bartok et al. also reported inaccurate predictions for % dehydration and various blood measures in 25 male wrestlers [10]. However, similar to a study by Baker and colleagues [34], the assumptions of the model were not discussed, and R^2 was not reported. As demonstrated in the present study, there was a large change in the mean pre- and post-dehydration (Table 1), but the degree to which BML explains these changes (R^2) is equally important, especially when BML is assumed to be reflected by hydration biomarker scores. In addition to means and correlation

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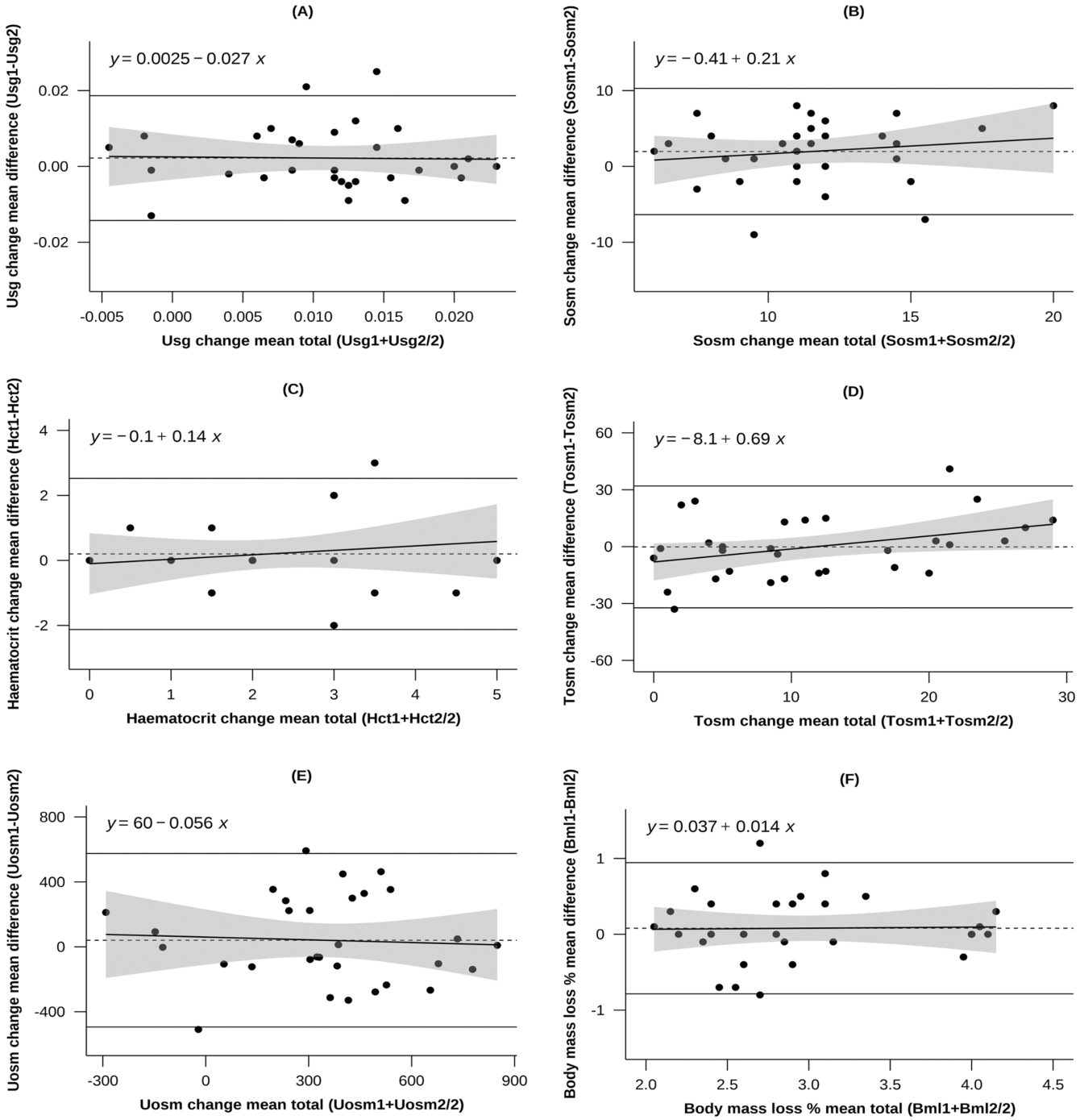

FIGURE 3 | Bland–Altman plots A–F illustrate the relationship between the average of the estimates from Trials 1 and 2 (x-axis) and the mean differences between Trials 1 and 2 (y-axis). The horizontal solid lines indicate the upper and lower 95% limits of agreement ([LOA], 1.96 ± SD), and the horizontal dashed line indicates the mean difference between trials. The regression line and equations are also displayed, where the gray shaded area represents the standard error for regression. Tear osmolarity was the only marker that showed a significant positive proportional bias ($p = 0.048$) between Trials 1 and 2.

testing, future research should consider regression analysis and provide comprehensive reporting of model results.

4.2 | Urinary Measures

Urinary measures are typically used to reflect hydration status in the field, due to their practicality, ease of use, and limited cost [9, 15]. However, the explanatory power of BML was lowest for the urinary biomarkers. Some studies have reported that urinary measures are sensitive to progressive dehydration [11, 12, 17], but correlation or regression analyses with BML were not conducted. Our study revealed similar mean differences from pre- to post-dehydration in T1 and T2, indicating a consistent response to dehydration at the group level. Like the blood analyses, correlation and regression

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TABLE 4 | Contingency tables for postdehydration values from both trials ($n = 60$).

(A)	(Impending dehydration) Serum osmolality <295 mOsm/kg	(Impending dehydration) Serum osmolality ≥295 mOsm/kg	(B)	(Dehydrated) Serum osmolality <301 mOsm/kg	(Dehydrated) Serum osmolality ≥301 mOsm/kg
Urine-specific gravity			Urine-specific gravity		
<1.020	True negative ($n = 4$) 6.6%	False negative ($n = 21$) 35%	<1.020	True negative ($n = 22$) 37%	False negative ($n = 3$) 5%
≥1.020	False positive ($n = 1$) 1.6%	True positive ($n = 34$) 57%	≥1.020	False positive ($n = 20$) 33%	True positive ($n = 15$) 25%
Total	5	55	Total	42	18
	True negative rate	True positive rate		True negative rate	True positive rate
	Specificity 80%	Sensitivity 62%		Specificity 52%	Sensitivity 83%

Note: Sensitivity = true positives/(true positives + false negatives). Specificity = true negatives/(true negatives + false positives).

analysis revealed a poor relationship between BML and urinary indices, suggesting a large variation in the individual response to acute dehydration. Indeed, only 35/60 USG samples were ≥1.020 postdehydration, which would result in 25 misclassifications in practice. However, the typical USG cutoff may be less sensitive to mild–moderate dehydration. Previous research has shown fewer false positive results as BML increases from 3% to 5% dehydration (58% and 33%), respectively [12]. Despite significant test–retest reliability, the ICCs were generally poor-moderate for the Uosm and USG measures, consistent with findings from Zubac et al. [13]. In contrast, Hamouti et al. found USG to be a reliable measure and suggested that adjustments for lean body mass may be needed [39]. However, tighter experimental controls, sample sizes, and protocol dissimilarities largely explain these results. The mean difference between T1 and T2 for USGΔ and UosmΔ was relatively low, but the SD was large, reflected in the wide LOA for both USG (−0.014 to 0.019) and Uosm (−494 to 575 mOsm/kg). Therefore, the explanatory power of BML and test–retest reliability of urinary biomarkers lack the desired precision for hydration testing in practice.

Studies have suggested that the first-morning USG is the most reliable urinary measure for assessing daily water balance [9], as the overnight equilibration of fluid compartments minimizes several potential confounding factors [6]. However, within the field, athletes are likely not weighing in before their first urination of the day. Even in rare cases where they might, athletes would likely prefer to empty their bladders and minimize dehydration, making it unrealistic to obtain a first-morning sample. Interestingly, the pretrial fluid bolus may have confounded some posturinary samples, but while necessary to establish euhydration, we found 7/60 negative USGΔ measurements. For example, even with 4% dehydration in T1 (87 min) and T2 (96 min), one participant showed negative USGΔ values of −0.007 and −0.002, respectively. As a result, athletes may be able to manipulate urine tests in regulatory settings, as evidenced by the unintentional decrease in USG. Other confounding factors and challenges with spot urinary testing have been previously described by Zubac et al. [9] and Cheuvront et al. [6] Despite previous research and the data presented here, spot samples are used to assess the precompetition hydration status. Given the numerous confounding factors and the variability in urine measurements, we argue that hydration testing via USG should not be used in a regulatory context.

4.3 | Tear Osmolarity

Tear osmolarity is a noninvasive, rapid, and convenient measure for assessing hydration status. However, similar to blood and urinary measures, BML was a poor predictor of TosmΔ, and some negative change scores (8/60) were observed. This is not a unique finding, with previous research reporting that some individuals experience a decrease in Tosm following dehydration [29, 40]. The exact reason for this is unclear. However, in the present study, it is possible that there was evaporation of tear fluid during the dehydration protocol and potential interaction with sweat, which may have affected the Tosm measurements. However, when post-Tosm was examined, which contains no negative values, a significant correlation was observed (Figure 2C). This finding is consistent with previous research reporting significant correlations between BML and post-Tosm [31, 40], suggesting that post-Tosm performs better than TosmΔ. Despite this, a larger variation is seen in postdehydration measures compared with baseline [18, 29, 31, 40]. In addition, the variation is reflected in the poor-moderate test–retest reliability for the Tosm measures, with the lowest ICCs of all variables for the TosmΔ and post-Tosm tests. Similarly, Ungaro et al. reported greater agreement before exercise (ICC = 0.79) than after exercise (ICC = 0.67). However, our analysis used single values, whereas Ungaro et al. assessed average measures, which typically have higher ICCs [33]. The Bland–Altman analysis and the fitted regression line identified systematic bias across trials, which is mirrored by the poor test–retest score for TosmΔ. Consistent with blood and urinary LOA, Tosm shows relatively large test–retest differences and has a high probability of misclassification if used in practice. The Tosm marker has only recently been investigated as a marker of hydration status, and while substantial interindividual variation is observed, confounding factors have yet to be confirmed. Future research should investigate potential confounders of Tosm and sample both eyes to increase confidence in the measure, as suggested by Ungaro [29]. Nevertheless, current data on Tosm suggest that it is too variable to be considered useful as a field measure.

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4.4 | Limitations

Although these findings hold significant value for hydration testing in combat sports, this study is not without limitations. We achieved a BML of only 3% on average, whereas in practice, these values in extreme cases can exceed 10%, limiting the generalizability of our findings to athletes who reduce BM by 2.0%–4.3%. Typically, athletes use a combination of methods to reduce BM, we only implemented passive dehydration and thus our results can only be extrapolated to when athletes dehydrate via sauna exclusively. Throughout both trials, the mean BML remained consistent. Despite careful monitoring, some participants displayed variations in sweat rate and volume, resulting in minor discrepancies in BML. However, athletes typically undergo similar but not identical weight cuts in preparation for weigh-ins, so these unintentional small differences in BML mirror real-world scenarios. Additionally, as BML increases so should biomarker values, which we did not observe with any degree of precision. Importantly, it is common practice for some athletes to engage in water loading, which in some cases may result in euhydration before their final water cut [5]. This aspect adds to the external validity of the study. However, not all athletes use water-loading or would be euhydrated, which also adds to the complexity of assessing biomarkers in the field. Despite these limitations, this is the first study to comprehensively evaluate multiple biomarkers following sauna-induced dehydration in both male and female combat sports athletes.

4.5 | Perspectives

Since 1998, hydration testing has been a part of the minimum weight program in wrestling. Recently, organizations have turned to USG tests at weigh-ins to assess athlete hydration levels. In our study, utilizing a dehydration method common among combat sports athletes, we evaluated the accuracy of BML in explaining changes in blood, urine, and tear markers, and their test–retest reliability. Our findings revealed that both accuracy and reliability fall short of the precision needed, even in a laboratory setting. We acknowledge that factors beyond BML, such as diet, exercise, individual physiology, biological variation, and the method and magnitude of dehydration, likely significantly contribute to changes in hydration biomarkers. In the field, where various factors, including fluid, exercise, alcohol, and caffeine intake, are not controlled for, even greater participant variation is expected. This renders them unsuitable for regulatory use and, if used, may result in an unfair advantage or disadvantage for some athletes. Based on previous research, the data presented here, and the unpredictable nature of how athletes will dehydrate and to what extent, alternative solutions should be investigated to manage weight cutting in these sports. Furthermore, our findings are pertinent not only to weight class sports but also to hydration testing across various contexts, including occupational settings, future research endeavors, and other sporting disciplines.

5 | Conclusion

Our findings revealed a wide range of biomarker scores between participants, even under conditions of weight stability at baseline. Blood parameters were most sensitive to BML, but only 11%–22% of biomarker variation was explained. The test–retest reliability was generally poor-moderate, and TosmΔ and post-Tosm had the lowest ICCs. Additionally, the selected cutoff values for Sosm and USG resulted in the misclassification of several dehydrated athletes, with the 301 mOsm/kg Sosm cutoff demonstrating greater sensitivity than the 295 mOsm/kg cutoff. In particular, if the USG 1.020 cutoff was implemented, our data suggest 42% of the athletes would be misclassified postdehydration. Consequently, BML should not be expected to accurately explain biomarker scores following sauna-induced dehydration. This study highlights the challenges associated with accurately assessing hydration status in a laboratory setting. We emphasize that the regulatory use of hydration testing is likely, in some cases, to lead to unjust disqualification or the inclusion of competitors, potentially adversely impacting their health, careers, and often challenging financial situations.

Acknowledgments

The authors would like to thank all participants for their time and effort. This study was designed by C.S.D., O.R.B., and L.V.F. The data were collected and analyzed by C.S.D. All authors contributed to manuscript preparation and approved the final version of this paper. The authors thank Dr. Chandra Salgado Kent for her assistance in the data analysis. Open access publishing facilitated by Edith Cowan University, as part of the Wiley - Edith Cowan University agreement via the Council of Australian University Librarians.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets are available from the corresponding author upon reasonable request.

References

1. O. R. Barley, D. W. Chapman, and C. R. Abbiss, "Weight Loss Strategies in Combat Sports and Concerning Habits in Mixed Martial Arts," *International Journal of Sports Physiology and Performance* 13, no. 7 (2018): 933–939, https://doi.org/10.1123/ijspp.2017-0715.

2. A. M. Kasper, B. Crighton, C. Langan-Evans, et al., "Case Study: Extreme Weight Making Causes Relative Energy Deficiency, Dehydration, and Acute Kidney Injury in a Male Mixed Martial Arts Athlete," *International Journal of Sport Nutrition and Exercise Metabolism* 29, no. 3 (2019): 331–338, https://doi.org/10.1123/ijsnem.2018-0029.

3. G. G. Artioli, B. Saunders, R. T. Iglesias, and E. Franchini, "It is Time to Ban Rapid Weight Loss From Combat Sports," *Sports Medicine* 46, no. 11 (2016): 1579–1584, https://doi.org/10.1007/s40279-016-0541-x.

4. A. C. Utter, "The New National Collegiate Athletic Association Wrestling Weight Certification Program and Sport-Seasonal Changes in Body Composition of College Wrestlers," *The Journal of Strength and Conditioning Research* 15, no. 3 (2001): 296–301, https://doi.org/10.1519/1533-4287(2001)015<0296:TNNCAA>2.0.CO;2.

5. O. R. Barley, D. W. Chapman, and C. R. Abbiss, "Reviewing the Current Methods of Assessing Hydration in Athletes," *Journal of the International Society of Sports Nutrition* 17, no. 1 (2020): 52, https://doi.org/10.1186/s12970-020-00381-6.

6. S. N. Cheuvront, R. W. Kenefick, and E. J. Zambraski, "Spot Urine Concentrations Should Not Be Used for Hydration Assessment: A Methodology Review," *International Journal of Sport Nutrition and Exercise*

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Metabolism 25, no. 3 (2015): 293–297, https://doi.org/10.1123/ijsnem.2014-0138.

7. D. Zubac, U. Marusic, and H. Karninčič, "Hydration Hydration Status Assessment Techniques and Their Applicability Among Olympic Combat Sports Athletes: Literature Review," *Strength & Conditioning Journal* 38, no. 4 (2016): 80–89, https://doi.org/10.1519/SSC.0000000000000236.

8. D. Zubac, A. Paravlic, R. Reale, I. Jelaska, S. A. Morrison, and V. Ivancev, "Fluid Balance and Hydration Status in Combat Sport Olympic Athletes: A Systematic Review With Meta-Analysis of Controlled and Uncontrolled Studies," *European Journal of Nutrition* 58, no. 2 (2019): 497–514, https://doi.org/10.1007/s00394-019-01937-2.

9. D. Zubac, R. Reale, H. Karnincic, A. Sivric, and I. Jelaska, "Urine Specific Gravity as an Indicator of Dehydration in Olympic Combat Sport Athletes; Considerations for Research and Practice," *European Journal of Sport Science* 18, no. 7 (2018): 920–929, https://doi.org/10.1080/17461391.2018.1468483.

10. C. Bartok, D. A. Schoeller, J. C. Sullivan, R. R. Clark, and G. L. Landry, "Hydration Testing in Collegiate Wrestlers Undergoing Hypertonic Dehydration," *Medicine and Science in Sports and Exercise* 36, no. 3 (2004): 510–517, https://doi.org/10.1249/01.MSS.0000117164.25986.F6.

11. R. A. Oppliger, S. A. Magnes, L. R. A. Popowski, and C. V. Gisolfi, "Accuracy of Urine Specific Gravity and Osmolality as Indicators of Hydration Status," *International Journal of Sport Nutrition and Exercise Metabolism* 15, no. 3 (2005): 236–251, https://doi.org/10.1123/ijsnem.15.3.236.

12. L. A. Popowski, R. A. Oppliger, G. P. Lambert, R. F. Johnson, A. K. Johnson, and C. V. Gisolfi, "Blood and Urinary Measures of Hydration Status During Progressive Acute Dehydration," *Medicine and Science in Sports and Exercise* 33, no. 5 (2001): 747–753, https://doi.org/10.1097/00005768-200105000-00011.

13. D. Zubac, D. Cular, and U. Marusic, "Reliability of Urinary Dehydration Markers in Elite Youth Boxers," *International Journal of Sports Physiology and Performance* 13, no. 3 (2018): 374–381, https://doi.org/10.1123/ijspp.2016-0621.

14. D. Zubac, T. Antelj, D. Olujic, V. Ivancev, and S. A. Morrison, "Fluid Balance and Hydration Assessment During the Weight-Stable Preparation Phase in Elite Youth Boxers," *Journal of Sports Sciences* 35, no. 8 (2017): 719–726, https://doi.org/10.1080/02640414.2016.1184302.

15. L. M. Sommerfield, S. R. McAnulty, J. M. McBride, et al., "Validity of Urine Specific Gravity When Compared With Plasma Osmolality as a Measure of Hydration Status in Male and Female NCAA Collegiate Athletes," *Journal of Strength and Conditioning Research* 30, no. 8 (2016): 2219–2225, https://doi.org/10.1519/JSC.0000000000001313.

16. R. W. Kenefick, S. N. Cheuvront, L. D. Elliott, B. R. Ely, and M. N. Sawka, "Biological and Analytical Variation of the Human Sweating Response: Implications for Study Design and Analysis," *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology* 302, no. 2 (2012): R252–R258, https://doi.org/10.1152/ajpregu.00456.2011.

17. N. Hamouti, J. Del Coso, and R. Mora-Rodriguez, "Comparison Between Blood and Urinary Fluid Balance Indices During Dehydrating Exercise and the Subsequent Hypohydration When Fluid Is Not Restored," *European Journal of Applied Physiology* 113, no. 3 (2013): 611–620, https://doi.org/10.1007/s00421-012-2467-9.

18. J. A. Owen, M. B. Fortes, S. Ur Rahman, M. Jibani, N. P. Walsh, and S. J. Oliver, "Hydration Marker Diagnostic Accuracy to Identify Mild Intracellular and Extracellular Dehydration," *International Journal of Sport Nutrition and Exercise Metabolism* 29, no. 6 (2019): 604–611, https://doi.org/10.1123/ijsnem.2019-0022.

19. L. E. Armstrong, C. M. Maresh, J. W. Castellani, et al., "Urinary Indices of Hydration Status," *International Journal of Sport Nutrition* 4, no. 3 (1994): 265–279, https://doi.org/10.1123/ijsn.4.3.265.

20. C. X. Muñoz, E. C. Johnson, J. K. Demartini, et al., "Assessment of Hydration Biomarkers Including Salivary Osmolality During Passive and Active Dehydration," *European Journal of Clinical Nutrition* 67, no. 12 (2013): 1257–1263, https://doi.org/10.1038/ejcn.2013.195.

21. R. J. Maughan, S. M. Shirreffs, and J. B. Leiper, "Errors in the Estimation of Hydration Status From Changes in Body Mass," *Journal of Sports Sciences* 25, no. 7 (2007): 797–804, https://doi.org/10.1080/02640410600875143.

22. S. N. Cheuvront and S. J. Montain, "Myths and Methodologies: Making Sense of Exercise Mass and Water Balance," *Experimental Physiology* 102, no. 9 (2017): 1047–1053, https://doi.org/10.1113/EP086284.

23. S. H. Mendes, A. C. Tritto, J. P. L. P. Guilherme, et al., "Effect of Rapid Weight Loss on Performance in Combat Sport Male Athletes: Does Adaptation to Chronic Weight Cycling Play a Role?" *British Journal of Sports Medicine* 47, no. 18 (2013): 1155–1160, https://doi.org/10.1136/bjsports-2013-092689.

24. L. E. Armstrong, R. J. Maughan, L. C. Senay, and S. M. Shirreffs, "Limitations to the Use of Plasma Osmolality as a Hydration Biomarker," *The American Journal of Clinical Nutrition* 98, no. 2 (2013): 502–512, https://doi.org/10.3945/ajcn.113.065466.

25. S. N. Cheuvront, R. W. Kenefick, N. Charkoudian, and M. N. Sawka, "Physiologic Basis for Understanding Quantitative Dehydration Assessment," *The American Journal of Clinical Nutrition* 97, no. 3 (2013): 455–462, https://doi.org/10.3945/ajcn.112.044172.

26. S. N. Cheuvront, B. R. Ely, R. W. Kenefick, and M. N. Sawka, "Biological Variation and Diagnostic Accuracy of Dehydration Assessment Markers," *American Journal of Clinical Nutrition* 92, no. 3 (2010): 565–573, https://doi.org/10.3945/ajcn.2010.29490.

27. V. E. Fernández-Elías, A. Martínez-Abellán, J. M. López-Gullón, et al., "Validity of Hydration Non-Invasive Indices During the Weight-cutting and Official Weigh-In for Olympic Combat Sports," *PLoS ONE* 9, no. 4 (2014): e95336, https://doi.org/10.1371/journal.pone.0095336.

28. T. D. Hew-Butler, C. Eskin, J. Bickham, M. Rusnak, and M. Vandermeulen, "Dehydration is How You Define It: Comparison of 318 Blood and Urine Athlete Spot Checks," *BMJ Open Sport & Exercise Medicine* 4, no. 1 (2018): 3–9, https://doi.org/10.1136/bmjsem-2017-000297.

29. C. T. Ungaro, A. J. Reimel, R. P. Nuccio, K. A. Barnes, M. D. Pahnke, and L. B. Baker, "Non-Invasive Estimation of Hydration Status Changes Through Tear Fluid Osmolarity During Exercise and Post-Exercise Rehydration," *European Journal of Applied Physiology* 115, no. 5 (2015): 1165–1175, https://doi.org/10.1007/s00421-015-3099-7.

30. S. N. Cheuvront and R. W. Kenefick, "Dehydration: Physiology, Assessment, and Performance Effects," *Comprehensive Physiology* 4, no. 1 (2014): 257–285, https://doi.org/10.1002/cphy.c130017.

31. M. B. Fortes, B. C. Diment, U. Di Felice, et al., "Tear Fluid Osmolarity as a Potential Marker of Hydration Status," *Medicine and Science in Sports and Exercise* 43, no. 8 (2011): 1590–1597, https://doi.org/10.1249/MSS.0b013e31820e7cb6.

32. G. E. W. Giersch, A. T. Colburn, M. C. Morrissey, et al., "Effects of Sex and Menstrual Cycle on Volume-Regulatory Responses to 24-h Fluid Restriction," *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology* 319, no. 5 (2020): R560–R565, https://doi.org/10.1152/ajpregu.00173.2020.

33. T. K. Koo and M. Y. Li, "A Guideline of Selecting and Reporting Intraclass Correlation Coefficients for Reliability Research," *Journal of Chiropractic Medicine* 15, no. 2 (2016): 155–163, https://doi.org/10.1016/j.jcm.2016.02.012.

34. L. B. Baker, J. A. Lang, and K. W. Larry, "Change in Body Mass Accurately and Reliably Predicts Change in Body Water After Endurance Exercise," *European Journal of Applied Physiology* 105, no. 6 (2009): 959–967, https://doi.org/10.1007/s00421-009-0982-0.

35. C. Ring, S. M. Patterson, S. L. Bacon, J. J. C. S. Veldhuijzen van Zanten, G. Willemsen, and D. Carroll, "Reliability of Hematocrit

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During Rest and Stress in Healthy Adults," *Biological Psychology* 77, no. 1 (2008): 63–68, https://doi.org/10.1016/j.biopsycho.2007.09.005.

36. O. R. Barley, D. W. Chapman, G. Mavropalias, and C. R. Abbiss, "The Influence of Heat Acclimation and Hypohydration on Post-Weight-Loss Exercise Performance," *International Journal of Sports Physiology and Performance* 15, no. 2 (2020): 213–221, https://doi.org/10.1123/ijspp.2019-0092.

37. K. J. Sollanek, R. W. Kenefick, S. N. Cheuvront, and R. S. Axtell, "Potential Impact of a 500-Ml Water Bolus and Body Mass on Plasma Osmolality Dilution," *European Journal of Applied Physiology* 111, no. 9 (2011): 1999–2004, https://doi.org/10.1007/s00421-011-1833-3.

38. A. C. Utter, S. R. McAnulty, B. F. Riha, B. A. Pratt, and J. M. Grose, "The Validity of Multifrequency Bioelectrical Impedance Measures to Detect Changes in the Hydration Status of Wrestlers During Acute Dehydration and Rehydration," *Journal of Strength and Conditioning Research* 26, no. 1 (2012): 9–15, https://doi.org/10.1519/JSC.0b013e318238ea51.

39. N. Hamouti, C. J. Del, A. Ávila, and R. Mora-Rodriguez, "Effects of Athletes' Muscle Mass on Urinary Markers of Hydration Status," *European Journal of Applied Physiology* 109, no. 2 (2010): 213–219, https://doi.org/10.1007/s00421-009-1333-x.

40. J. J. Holland, M. Ray, C. Irwin, T. L. Skinner, M. Leveritt, and B. Desbrow, "Tear Osmolarity is Sensitive to Exercise-Induced Fluid Loss But is Not Associated With Common Hydration Measures in a Field Setting," *Journal of Sports Sciences* 36, no. 11 (2018): 1220–1227, https://doi.org/10.1080/02640414.2017.1365157.

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